Multimedia Signal Processing - 2nd Module

July 20, 2023

1. **Question (7 pts)**
   Consider two sources X and Y that emit symbols according to the following joint p.m.f.

	x_1	x_2	x_3	x_4
y_1	?	1/8	1/8	1/8
y_2	1/16	1/4	1/4	1/16

   (a) Define the value of "?" in order to obtain a valid p.m.f.

   (b) Define the alphabet of the source X, the alphabet of the source Y, and their cardinality.

   (c) Estimate the probability mass functions $p_X(x)$ of X and $p_Y(y)$ of Y.

   (d) Compute the entropy $H(X)$ and the entropy $H(Y)$.

   (e) Consider the joint source (X, Y) emitting one symbol from X and one symbol from Y at each time instant. Compute the Huffman code, its length, and compare it to the theoretical bound.

   (f) Compute the extension code for the series of symbols $[(x_1, y_2), (x_1, y_2), (x_4, y_1)]$.

   (g) Can you write the extension code for the series of symbols $[(x_1, y_2), (x_1, y_1), (x_3, y_1)]$? Why?

2. **Question (5 pts)** Consider the problem of video coding.

   (a) Draw the block diagram of a typical hybrid DCT/DPCM video codec.

   (b) Describe each block.

   (c) Describe the concept of displaced frame difference (DFD).

3. **Question (3 pts)** Consider the problem of lossless coding two discrete memoryless sources X and Y.

   (a) Provide the mathematical definition of entropy $H(X)$ starting from the definition of information of a symbol.

   (b) Describe the meaning of the entropy $H(X)$.

   (c) Define how it is possible to compute the joint entropy $H(X, Y)$ if you only know $H(X)$, $H(Y)$, $H(X|Y)$ and the mutual information $I(X, Y)$.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider the following codebook describing three codewords: $C_1 = 0$, $C_2 = 1$, $C_3 = 0$. This codebook defines a binary code.
   - [T] [F] The Optimum Uniform Scalar Quantizer provides the lowest possible MSE at the same rate compared to other scalar quantizers.
   - [T] [F] Consider a discrete memoryless source X with the following alphabet $\mathcal{A} = \{\alpha, \beta, \gamma\}$. This source can have entropy $H(X) = 2$.
   - [T] [F] Consider a transform coding scheme that maps each input vector $\mathbf{x} = [x_1, x_2]$ into a vector of coefficients $\mathbf{y} = [y_1, y_2]$. The sum of the variances of x_1 and x_2 is equal to the sum of the variances of the transformed coefficients y_1 and y_2.
   - [T] [F] In a video coding scheme, motion vectors can represent motion in terms of fraction of pixels.
   - [T] [F] Audio tonal masking only happens at some specific frequencies.
   - [T] [F] Consider two discrete sources X and Y. The mutual information can be $I(X, Y) = 0$.
   - [T] [F] Consider the problem of synthesizing speech voiced sounds using the source-filter model and a train of pulses as input to the filter. The distance between consecutive pulses affects the pitch of the synthesized voiced sound.

5. **Question (Matlab - 10pts)**

   Load the build-in 'pears.png' image file using the command `img = imread('pears.png')`. Now the variable `img` contains a 3D matrix (i.e., height x width x color) representing an RGB color image. This is your input data.

   (a) Crop the top-left 128×128 pixels of the image and display it in full color (use the cropped version hereinafter).

   (b) Extract only the red component of the cropped image (use the red component only hereinafter).

   (c) Encode and decode the cropped red component using a 4x4 DCT block-wise transform coding technique. In doing so, consider the following steps:

   - Compute how many 4x4 blocks can be extracted.
   - Define the quantizer to be used. The quantizer should use 16 symbols uniformly distributed (*hint*: pay attention to the quantizer domain).
   - Compute the encoded version of the image.
   - Compute the decoded version of the image.

   (d) Compute the achieved rate (use entropy as measure of rate).

   (e) You have used a single quantizer for all the components. Explain if this is a good idea in transform coding or not.